

Assignment 6

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Econometrics I
NYU Stern

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Instructions. Two problems on regression discontinuity and synthetic control (Sessions 10–11). R with `fixest` or Python with `pyfixest`. For the synthetic control you may use a package (`Synth`, `tidysynth`, or `synthdid` in R; `pysyncon` in Python) or write the eight-line version described in Problem 2. Turn in a PDF with your results and reasoning, and your code.

1 Rating rounding: sharp, then fuzzy

Based on Anderson and Magruder (2012), who use Yelp’s half-star rounding. A platform computes each restaurant’s average rating r_i to two decimals but *displays* it rounded to the nearest half star. Restaurants with $r_i \geq 3.75$ display four stars; those just below display three and a half. Simulate 3,000 restaurants with $r_i \sim \text{Uniform}(3.25, 4.25)$, let $x_i = r_i - 3.75$ be the distance to the threshold, and let log weekly revenue be

$$y_i = 10 + 1.5x_i + 0.8x_i^2 + 0.6 \cdot \mathbf{1}\{x_i \geq 0\} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

Underlying quality matters smoothly; the displayed star adds a jump of 0.6 at the threshold.

- (a) Naive comparison: mean y for restaurants displaying four stars minus those displaying three and a half. Why is this not the effect of the extra half star?
- (b) Binscatter: mean y in 20 equal-width bins of x ; plot with a vertical line at zero.
- (c) Local linear RDD: regress y on $\mathbf{1}\{x \geq 0\}$, x , and their interaction, using restaurants within h of the threshold, for $h = 0.5, 0.25$, and 0.10 . Report the jump and its robust standard error at each bandwidth. What is the tradeoff as h shrinks, and how does the $0.8x^2$ term in the truth show up in it?

- (d) Placebo cutoffs at $x = -0.25$ and $x = +0.25$, same bandwidth 0.25. What should you find, and what do you find?
- (e) Manipulation check: compare the number of restaurants just below and just above the threshold (within 0.05). Explain in one paragraph what a pile-up just above would suggest in the real setting, and why it would invalidate the design.
- (f) Fuzzy version. Now the displayed star does not move revenue directly; instead it raises the chance the platform *features* the restaurant. Simulate an unobserved quality $q_i \sim \mathcal{N}(0, 1)$ and

$$F_i = \mathbf{1}\{-0.8 + 1.4 \cdot \mathbf{1}\{x_i \geq 0\} + 0.8 q_i + \nu_i > 0\}, \quad y_i = 10 + 1.5 x_i + 0.8 x_i^2 + 1.0 F_i + 0.8 q_i + \varepsilon_i,$$

with $\nu_i, \varepsilon_i \sim \mathcal{N}(0, 1)$. The effect of being featured is 1.0. Within $h = 0.25$: (i) regress y on F (and the running-variable terms); (ii) estimate the reduced form and the first stage at the threshold; (iii) form the Wald ratio and compare to 2SLS with the threshold dummy as the instrument for F . Why does (i) fail and (iii) work, and for which restaurants is the 2SLS number the effect of being featured?

2 A soda tax, one city, thirty donors

A city imposes a soda tax at the start of quarter 13 of a 20-quarter panel; 30 other cities do not. Per-capita soda sales follow a factor structure,

$$Y_{jt} = 100 + 5(\lambda_{1t} \mu_{j1} + \lambda_{2t} \mu_{j2}) + \varepsilon_{jt}, \quad \varepsilon_{jt} \sim \mathcal{N}(0, 1),$$

with λ_{1t} a linear trend from 0 to 2 over the 20 quarters, $\lambda_{2t} \sim \mathcal{N}(0, 1)$ i.i.d. over t , and city loadings $\mu_{j1}, \mu_{j2} \sim \text{Uniform}(0, 1)$ for the donors. Give the treated city loadings (0.9, 0.5): it is a fast-growing city. The tax cuts its sales by 10% in every post-tax quarter (multiply Y_{1t} by 0.9 for $t \geq 13$).

- (a) Simulate. Plot the treated city against the average of the donors. Estimate the effect by difference-in-differences with all donors as the control group. Why is it biased here, and in which direction?
- (b) Build the synthetic control: donor weights $w \geq 0$, $\sum_j w_j = 1$, chosen to minimize the pre-period squared gap $\sum_{t \leq 12} (Y_{1t} - \sum_j w_j Y_{jt})^2$. (Eight-line version: parameterize $w_j = e^{\theta_j} / \sum_k e^{\theta_k}$ and hand the objective to `optim`; or use `quadprog::solve.QP`; or a package.) Report the weights above 0.05, the pre-period RMSPE, and the average post-period gap. Compare to the truth and to (a).
- (c) Why does the synthetic control get this right when DiD does not? Answer in terms of the factor loadings.

- (d) Placebo in space: give the treatment to each donor in turn (with the treated city removed from the donor pool), compute each donor's ratio of post- to pre-period RMSPE, and place the treated city's ratio in that distribution. Report the implied p -value. What does this test assume?
- (e) Now move the treated city's loadings to $(1.3, 0.5)$, outside the range of the donors. Re-run (b). What happens to the pre-period fit, and what does that tell you about when synthetic control can and cannot be used?